

Question	Answer	Marks
4(a)	random/haphazard	B1
	constant velocity or speed in a straight line between collisions or distribution of speeds/different directions	B1
4(b)	(small) specks of light/bright specks/pollen grains/dust particles/smoke particles	M1
	moving haphazardly/randomly/jerky/in a zigzag fashion	A1
4(c)(i)	$pV = \frac{1}{3} Nm\langle c^2 \rangle$ $1.05 \times 10^5 \times 0.0240 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2 \rangle$	C1
	$\langle c^2 \rangle = 1.89 \times 10^6$	C1
	or	
	$\frac{1}{2} m\langle c^2 \rangle = (3/2) kT$ $0.5 \times (4.00 \times 10^{-3} / 6.02 \times 10^{23}) \times \langle c^2 \rangle = 1.5 \times 1.38 \times 10^{-23} \times 300$	(C1)
	$\langle c^2 \rangle = 1.87 \times 10^6$	(C1)
	or	
	$nRT = \frac{1}{3} Nm\langle c^2 \rangle$ $1.00 \times 8.31 \times 300 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2 \rangle$	(C1)
	$\langle c^2 \rangle = 1.87 \times 10^6$	(C1)
	$c_{r.m.s.} = 1.37 \times 10^3 \text{ m s}^{-1}$	A1

Question	Answer	Marks
4(c)(ii)	$\langle c^2 \rangle \propto T$	C1
	$\langle c^2 \rangle$ at 177 °C = $1.89 \times 10^6 \times (450 / 300)$	C1
	$c_{\text{r.m.s.}}$ at 177 °C = $1.68 \times 10^3 \text{ m s}^{-1}$	A1

Question	Answer	Marks
5(a)	(loss in) kinetic energy of α particle = $Q\alpha / 4\pi\epsilon_0 r$	C1